

201. Ferrous sulphate on strong heating gives

- (a) SO_2
- (b) $\text{Fe}_2(\text{SO}_4)_3$
- (c) $\text{FeO} + \text{SO}_3$
- (d) $\text{Fe}_2\text{O}_3 + \text{SO}_2 + \text{SO}_3$

Answer: D — (d) $\text{Fe}_2\text{O}_3 + \text{SO}_2 + \text{SO}_3$

Chemical reason: Applying the known reaction of the stated transition-metal compound under these reagents/conditions gives $\text{Fe}_2\text{O}_3 + \text{SO}_2 + \text{SO}_3$. The choice is consistent with the expected oxidation state and stability of the product.

202. Green vitriol is

- (a) $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$
- (b) $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$
- (c) $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$
- (d) $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$

Answer: B — (b) $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$

Chemical reason: This is a standard nomenclature/composition identification: the name or formula given in the question corresponds to $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$. The choice is fixed by the known composition of that compound/mineral.

203. When conc. sulphuric acid is added slowly to a solution of ferrous sulphate containing nitrate

ion, a brown colour ring is formed. The composition of the ring is

- (a) $[\text{Fe}(\text{H}_2\text{O})_5\text{NO}]\text{SO}_4$
- (b) $\text{FeSO}_4 \cdot \text{NO}_2$
- (c) $\text{Fe}[(\text{H}_2\text{O})_5](\text{NO}_3)_2$
- (d) None of these

Answer: A — (a) $[\text{Fe}(\text{H}_2\text{O})_5\text{NO}]\text{SO}_4$

Chemical reason: The colour follows from the electronic structure and coordination environment of the transition-metal ion. The species in $[\text{Fe}(\text{H}_2\text{O})_5\text{NO}]\text{SO}_4$ has the characteristic oxidation state/ligand field that produces the observed colour.

204. F_2 is formed by reacting K_2MnF_6 with

- (a) SbF_5
- (b) MnF_3
- (c) KSbF_6
- (d) MnF_4

Answer: A — (a) SbF_5

Chemical reason: Applying the known reaction of the stated transition-metal compound under these reagents/conditions gives SbF_5 . The choice is consistent with the expected oxidation state and stability of the product.



205. Railway wagon axles are made by heating rods of iron embedded in charcoal powder. The process is known as

- (a) Case hardening
- (b) Sheradizing
- (c) Annealing
- (d) Tempering

Answer: A — (a) Case hardening

Chemical reason: Case hardening enriches the surface of iron/low-carbon steel with carbon by heating it in contact with carbonaceous material. The hard surface and tougher core are useful for axles and gears.

206. The alloy of steel that is used in making automobile parts and utensils

- (a) Stainless steel
- (b) Nickel steel
- (c) Tungstun steel
- (d) Chromium steel

Answer: A — (a) Stainless steel

Chemical reason: This is determined by the standard composition or heat-treatment behaviour of the metal/alloy. The description in Stainless steel matches the relevant metallurgical process or alloy property.

207. Which of the following has lowest percentage of carbon

- (a) Cast iron
- (b) Wrought iron
- (c) Steel
- (d) All have same percentage

Answer: B — (b) Wrought iron

Chemical reason: Wrought iron has the lowest carbon content, typically below about 0.1–0.15%, whereas steel and cast iron contain progressively more carbon.

208. Galvanisation is the

- (a) Deposition of Zn on Fe
- (b) Deposition of Al on Fe
- (c) Deposition of Sn on Fe
- (d) Deposition of Cu on Fe

Answer: A — (a) Deposition of Zn on Fe

Chemical reason: This is determined by the standard composition or heat-treatment behaviour of the metal/alloy. The description in Deposition of Zn on Fe matches the relevant metallurgical process or alloy property.

209. Tempered steel is

- (a) Soft and pliable
- (b) Hard and brittle
- (c) Neither so hard nor so brittle
- (d) Very soft

Answer: C — (c) Neither so hard nor so brittle

Chemical reason: Tempering partially reverses the extreme hardness/brittleness caused by quenching. Properly tempered steel is neither as hard nor as brittle as freshly hardened steel.



210. Best quality of steel is manufactured by

- (a) Siemen –Martin's open hearth process
- (b) Electrical process
- (c) Bessemer process
- (d) Blast furnace

Answer: A — (a) Siemen –Martin's open hearth process

Chemical reason: Electric-furnace steelmaking provides close temperature and composition control and can produce very high-purity, high-quality steels.

211. The presence of Si in steel gives it

- (a) Fibrous structure
- (b) Silicate type structure
- (c) Sheet type structure
- (d) None of these

Answer: A — (a) Fibrous structure

Chemical reason: This is determined by the standard composition or heat-treatment behaviour of the metal/alloy. The description in Fibrous structure matches the relevant metallurgical process or alloy property.

212. The presence of Mn in steel produces

- (a) Elasticity
- (b) Increases tensile strength
- (c) Both
- (a) and
- (b)
- (d) None of these

Answer: C — (c) Both (a) and (b)

Chemical reason: This is determined by the standard composition or heat-treatment

behaviour of the metal/alloy. The description in Both (a) and (b) matches the relevant metallurgical process or alloy property.

213. Presence of Cr in steel makes it

- (a) Resistant to chemical action
- (b) Useful for making cutlery
- (c) Increases chemical action
- (d)
- (a) and
- (b) both

Answer: A — (a) Resistant to chemical action

Chemical reason: This is determined by the standard composition or heat-treatment behaviour of the metal/alloy. The description in Resistant to chemical action matches the relevant metallurgical process or alloy property.

214. The addition of metals like Cr, Mn, W and Ni to ordinary steel makes it

- (a) More useful
- (b) Alters the properties of ordinary steel
- (c) Both
- (a) and
- (b)
- (d) None of these

Answer: C — (c) Both (a) and (b)

Chemical reason: This is determined by the standard composition or heat-treatment behaviour of the metal/alloy. The description in Both (a) and (b) matches the relevant metallurgical process or alloy property.



215. Stainless steel is non-corrosive. This character is more prominent in

- (a) Mn steel
- (b) Ordinary steel
- (c) Ti steel
- (d) All of these

Answer: C — (c) Ti steel

Chemical reason: This is determined by the standard composition or heat-treatment behaviour of the metal/alloy. The description in Ti steel matches the relevant metallurgical process or alloy property.

216. When little vanadium is mixed with steel, it becomes

- (a) More hard
- (b) More tensility
- (c) Both
- (a) and
- (b)
- (d) No effect

Answer: C — (c) Both (a) and (b)

Chemical reason: A small addition of vanadium forms hard carbides and refines the grain of steel, increasing both hardness/strength and toughness/tensile performance.

217. To obtain steel entirely free from sulphur and phosphorus, the process used is

- (a) Electrothermal process
- (b) Bessemer process
- (c) Open-hearth process
- (d) Duplex process

Answer: A — (a) Electrothermal process

Chemical reason: The electrothermal/electric process can remove sulfur and phosphorus very effectively and permits close control of composition, so it is used for very pure or special steels.

218. Stainless steel contains Cr

- (a) 14%
- (b) 5%
- (c) 50%
- (d) 2.5%

Answer: A — (a) 14%

Chemical reason: This is determined by the standard composition or heat-treatment behaviour of the metal/alloy. The description in 14% matches the relevant metallurgical process or alloy property.

219. The chief constituents of steel made in India are

- (a) Mn and Cr
- (b) Al and Zn
- (c) V and Co
- (d) Ni and Mg

Answer: A — (a) Mn and Cr

Chemical reason: This is determined by the standard composition or heat-treatment





behaviour of the metal/alloy. The description in Mn and Cr matches the relevant metallurgical process or alloy property.

220. Which of the following is used to prepare medical instruments

- (a) Cast iron
- (b) Wrought iron
- (c) Steel
- (d) Alloy of Cu and Fe

Answer: C — (c) Steel

Chemical reason: The application follows from the characteristic chemical property of the material—such as catalytic activity, stability, conductivity, or corrosion resistance. That property is specifically associated with Steel.

